TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA International Joint R&D Program
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Joint validation and safety qualification of high-silicon composite and 3D-structured lithium-metal anodes using advanced diagnostics and model-based battery management			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics (Code 200901)		Electrical & Electronics (Code 200905)	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Battery Materials	Batteries	Secondary Batteries
Concept of Proposed Technology	SME develops next-generation anodes based on (i) high-silicon composite anodes and (ii) thin, 3D-structured lithium-metal foils. Key technical risks are mechanical and interfacial degradation in silicon (large volume change during cycling) [5] and non-uniform Li plating that can trigger dendrites and short circuits [6]. TNO proposes a joint R&D programme that combines: - Climate-controlled electrochemical validation and accelerated ageing using TNO's battery laboratory infrastructure, including a 300 kW battery cycler, coolant heating and cooling control, multi-channel ageing characterisation, and Electrochemical Impedance Spectroscopy (EIS) [1].			

	- Safety-relevant boundary testing in safe, inert environments with gas detection, and definition of safe operating envelopes for high-silicon and lithium-metal anodes. Laboratory equipment is Hardware-in-the-Loop (HiL) capable to validate control algorithms with representative hardware constraints [1,6]. - Model-based analysis and battery-management algorithm development (SoX estimation, charge control, diagnostics) building on TNO's BMS and digital-twin research, including work co-authored by Steven Wilkins [7,8,9]. - Evidence packaging aligned with the EU Battery Regulation (EU) 2023/1542 requirements for QR-code access and battery passport from 18 February 2027 [4] and TNO's Secure Battery Passport Demonstrator (SeBaPaD) activities [2,3]. The result is a validated, scalable pathway for SME to integrate advanced anodes into cells and modules with quantified performance, safety margins, and traceability-ready data.
Rationale for Joint R&D Project	SME;s anode innovations require system-level validation and degradation-aware operating strategies before scale-up. TNO contributes complementary capabilities: (i) a battery lab designed for performance validation of models and algorithms at cell, module, and pack level with HiL-capable equipment and EIS diagnostics [1]; (ii) published BMS and digital-twin methods (Wilkins et al.) to translate test evidence into robust control strategies [7,8,9]; and (iii) battery passport and data-trust tooling (SeBaPaD) to support EU market access from 2027 [2,4]. The joint project reduces technical and commercial risk by combining materials and prototype cells with TNO's validation, modelling, and compliance-oriented evidence framework.
Key Words	1. High-silicon composite anode 2. 3D host / structured Li foil 3. SEI stabilization/dendrite mitigation 4. Electromagnetic Impedance Spectroscopy (EIS) 5. Hardware-in-the-Loop

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	Start of the Project: lab-scale electrode and prototype cell validation	TRL 3-4	End of the Project: validated prototype in a relevant environment with operating envelope and diagnostics	TRL 6
R&D Objectives	1) Establish a joint test and modelling protocol for high-silicon and lithium-metal anodes (cells and representative half-cells) with agreed KPIs. 2) Generate a climate-dependent ageing and impedance dataset (EIS + standard cycling) to quantify degradation modes and parameterise models [1]. 3) Develop degradation-aware charging and operational strategies using model-based BMS and digital-twin methods (SoX estimation, charge profile prediction) [7,8]. 4) Define safety-relevant limits for lithium-metal plating stability and silicon mechanical degradation, and propose mitigation measures informed by 3D-host Li literature and test evidence [5,6]. 5) Produce a traceability-ready evidence package aligned with EU Battery Passport requirements and TNO SeBaPaD approach [2,4].			
R&D Target Specifications	Targets (to be demonstrated on agreed cell formats): - Silicon-composite anode: first-cycle coulombic efficiency >= 90% and capacity retention >= 80% after 300 cycles at 25 C (baseline defined in WP1). - Lithium-metal anode: average coulombic efficiency >= 99.0% in plating/stripping tests under agreed current density window, and no internal short in 200-cycle symmetric tests (conditions defined in WP1) [6]. - Diagnostics: EIS-based degradation indicators that correlate with capacity fade (R^2 >= 0.8) across temperature points [1]. - Control: a degradation-aware charging policy that reduces predicted degradation rate by >= 15% versus baseline, validated with the digital-twin workflow [8]. - Compliance: data fields and access control scheme sufficient to populate battery passport records per Regulation (EU) 2023/1542 (Article 77, Annex XIII) [4].			
R&D Plan	Work plan (36 months): WP1 (M1-M4) Requirements and baseline definition: cell formats, KPIs, safety envelope hypotheses, test matrix, data schema for passport. WP2 (M3-M14) Experimental validation: cycling across temperature setpoints, EIS campaigns, post-mortem selection and reporting [1]. WP3 (M4-M16) Modelling and algorithm development: parameter identification, degradation models, SoX estimation and charge control; integration with digital-twin workflows [7,8].			

	WP4 (M114-M28) Safety-relevant testing and mitigation: plating stability window, fast-charge stress tests; structured Li foil evaluation informed by 3D-host Li design criteria [6]. WP5 (M28-M36) Demonstration and evidence package: consolidated dataset, validated operating envelopes, documentation for battery passport readiness (QR code access, record structure) [4].

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO will: - Design and execute the validation campaign, including EIS and climate-dependent ageing, using HiL-capable equipment. The battery lab includes a 300 kW cycler, coolant heating and cooling control, multi-channel ageing equipment, and inert test environments with gas detection for secure testing [1]. - Lead model development and algorithm work (SoX estimation, charging policy, diagnostics) building on published BMS and digital-twin methods [7,8,9]. - Support safety-relevant test design and interpretation for lithium-metal and silicon degradation risks using current literature and lab evidence [5,6]. - Structure and deliver the evidence package aligned with EU Battery Passport requirements and the SeBaPaD reference architecture and algorithms [2,4]. - Provide project management, reporting and dissemination.
Expected R&R of Korean SME Partner	Korean SME will: - Provide anode materials, structured Li foils and prototype cells, including bill of materials and process windows required to interpret test results. - Define target application requirements (cell format, loading, current density limits, fast-charge profile constraints). - Implement design iterations (electrode architecture, coating, prelithiation strategy) based on joint test feedback. - Support integration into representative cell formats and coordinate scale-up and manufacturability assessment. - Participate in technical reviews, data governance decisions and joint IP planning.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes: - Validated performance and degradation dataset for SME's silicon-composite and lithium-metal anodes across temperature and load cases, including EIS signatures [1]. - A calibrated modelling and diagnostics workflow (digital-twin ready) and a degradation-aware charging strategy anchored in published methods [7,8,9]. - Defined operating envelopes and safety-relevant limits for advanced anodes, supported by literature and test evidence [5,6]. - Battery passport-ready evidence package consistent with Regulation (EU) 2023/1542 requirements from 18 February 2027 and aligned with TNO SeBaPaD activities [2,4]. Impact: - Faster de-risking and TRL increase for SME's anode technology towards automotive and industrial qualification. - Improved EU market readiness through traceability and compliance-oriented documentation. - Strengthened Netherlands-Korea collaboration in strategic secondary battery technology.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization